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Q1. If $p : q = r : s = t :$ [illegible] hat is the value of $(p + r + t) / (q + s + u)$?

A. 2 : 3

B. 3 : 8

C. 1 : 4

D. 4 : 5

$$\frac{p}{q} = \frac{r}{s} = \frac{t}{u} = \frac{2}{3}$$

$$\frac{p}{q} = \frac{2}{3} \Rightarrow \boxed{p = \frac{2q}{3}}$$

$$\frac{r}{s} = \frac{2}{3} \Rightarrow \boxed{r = \frac{2s}{3}}$$

$$\frac{t}{u} = \frac{2}{3} \Rightarrow \boxed{t = \frac{2u}{3}}$$

$$\frac{p+r+t}{q+s+u} = \frac{\frac{2q}{3} + \frac{2s}{3} + \frac{2u}{3}}{q+s+u}$$

$$= \frac{\frac{2}{3}[q+s+u]}{q+s+u}$$

$$= \frac{2}{3}$$

Ratio & Proportion

Q3. If a : b = c : d = 2 : 3 then what is the value of $(ab + cd) / (b^2 + d^2)$?

A. 2 : 3

B. 1 : 7

C. 3 : 2

D. 4 : 3

$$\frac{ab}{bb} = \frac{cd}{dd} = \frac{2}{3}$$

$$\frac{ab + cd}{b^2 + d^2} = \frac{2}{3}$$

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EQUIVALENCE RATIO PROPERTY

1. $\frac{a}{b} = \frac{c}{d} = \frac{a}{b} = \frac{c}{d} = \boxed{\frac{2a+3c}{2b+3d}} = \frac{ab+cd}{b^2+d^2}$, etc.

$\frac{a \cdot b}{b \cdot b} = \frac{c \cdot d}{d \cdot d}$

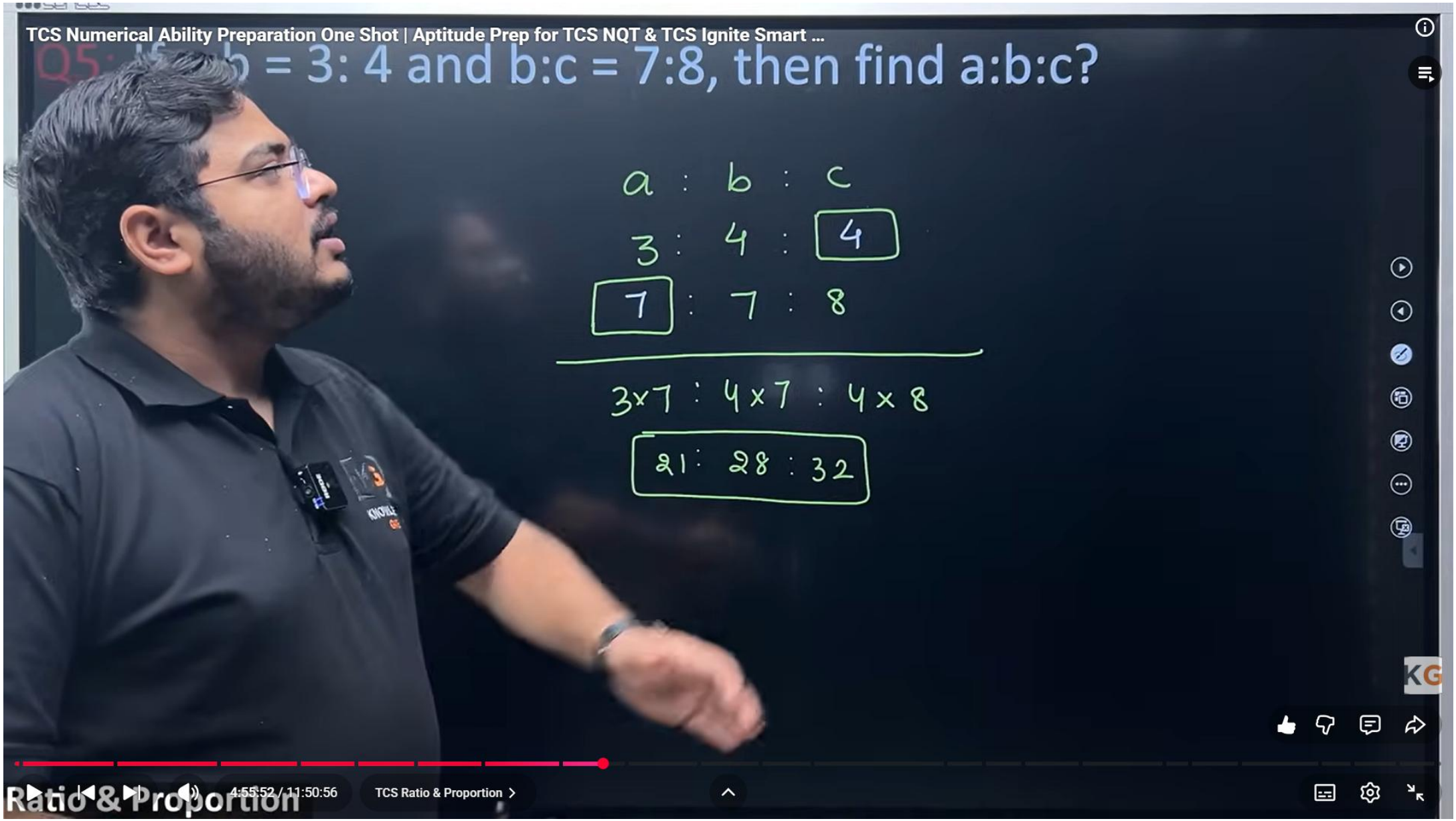
$\frac{ab+cd}{b^2+d^2}$

$\frac{a+c}{b+d}$

2. $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} \Rightarrow \frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{a+2c+3e}{b+2d+3f} = \frac{4a-3c+9e}{4b-3d+9f}$, etc.

$\frac{a+c+e}{b+d+f}$

4:51:39 / 11:50:56 TCS Ratio & Proportion
Ratio & Proportion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Q5: If a:b = 3: 4 and b:c = 7:8, then find a:b:c?
a : b : c
3 : 4 : 4
7 : 7 : 8
3×7 : 4×7 : 4×8
21 : 28 : 32
Ratio & Proportion
4:55:52 / 11:50:56
TCS Ratio & Proportion

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Q4: If a:b = 2: 3 and b:c = 5:6, then find a:b:c?

a : b : c

→ 2 : 3 : $\boxed{3}$

→ $\boxed{5}$: 5 : 6

10 : 15 : 18

a : b : c

2 : 3 : $\boxed{3}$

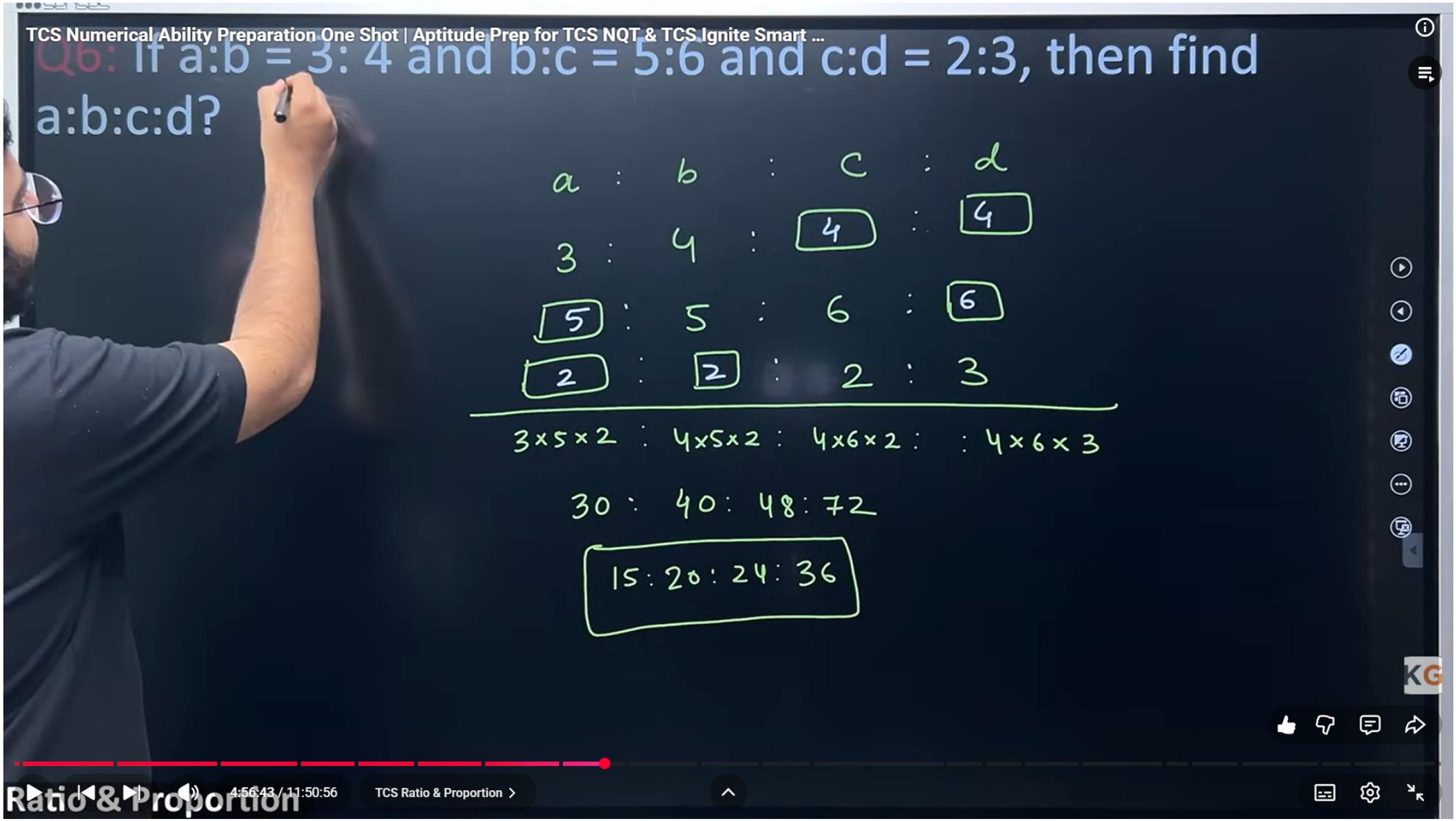
$\boxed{5}$: 5 : 6

2×5 : 3×5 : 3×6

$\boxed{10 : 15 : 18}$

$\boxed{a : b : c}$

Ratio & Proportion
4:55:15 / 11:50:56
TCS Ratio & Proportion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Q6: If a:b = 3: 4 and b:c = 5:6 and c:d = 2:3, then find a:b:c:d?
a : b : c : d
3 : 4 : 4 : 4
5 : 5 : 6 : 6
2 : 2 : 2 : 3
3×5×2 : 4×5×2 : 4×6×2 : : 4×6×3
30 : 40 : 48 : 72
15 : 20 : 24 : 36
4:56:43 / 11:50:56
TCS Ratio & Proportion
Ratio & Proportion

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7. If the ratio of x to y is 6/5 and the ratio of z to y is 3/4 then the ratio of x to z is

$\frac{x}{y} = \frac{6}{5}$

$\frac{z}{y} = \frac{3}{4} \Rightarrow \frac{y}{z} = \frac{4}{3}$

$\frac{x}{y} \cdot \frac{y}{z} = \frac{\cancel{6}^{2}}{5} \cdot \frac{4}{\cancel{3}}$

$\frac{x}{z} = \frac{8}{5}$

$x : z = 8 : 5$

M2

$x : y : z$

$6 : 5 : \boxed{5}$

$\boxed{4} : 4 : 3$

$6 \times 4 : 5 \times 4 : 5 \times 3$

$24 : 20 : 15$

$x : y : z$

$x : z = 24 : 15 = 8 : 5$

$\frac{24}{15} = \frac{8}{5}$

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Ratio & Proportion
4:59:47 / 11:50:56
TCS Ratio & Proportion

Q8. Divide Rs. 1162 among A, B, and C in the ratio 35 : 28 : 20.

M1

35 : 28 : 20 ⇒ sum ⇒ 35 + 28 + 20 = 83

$A = \frac{35}{83} \times 1162 = 35 \times 14 = \text{Rs. } 490$

$\frac{28}{83} \times 1162 = 28 \times 14 = \text{Rs. } 392$

$C = \frac{20}{83} \times 1162 = 20 \times 14 = \text{Rs. } 280$

Rs. 1162

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